Project Title

Smart Lender – Loan Eligibility Prediction System

Team Id : SWUID2026211879
SWUID2026196379
SWUID2026211883
SWUID2026211881
SWUID2026226496

Team Size: 5

Team Member : Kandala Srinadh

Team Member : Bheri Sri Ranga Satya Sai Kiran

Team Member : Rangana Nikhitha

Team Member : Puneeth Viswash Pasagada

Team Member : Ganesh Pavan Pedapudi

1. INTRODUCTION

1.1 Project Overview

The Smart Lender – Loan Eligibility Prediction System is a machine learning-based web application developed to predict whether a loan application is likely to be approved or rejected based on an applicant's financial and personal information. In the banking sector, loan approval is an important process that requires careful analysis of multiple factors before making a final decision. Traditional loan approval methods often involve manual verification, which can be time-consuming, labor-intensive, and prone to human errors. This project aims to simplify and automate that process using machine learning techniques.

The application collects important details from users, including gender, marital status, education, employment status, applicant income, co-applicant income, loan amount, loan term, credit history, and property area. These details are processed using a trained machine learning model that analyzes the data and predicts whether the applicant is eligible for a loan. The project is developed using Python, Flask, HTML, CSS, and Scikit-learn, providing an interactive and user-friendly web interface for real-time loan eligibility prediction.

The system helps reduce manual effort, improves prediction speed, and supports financial institutions in making faster and more consistent decisions. By integrating machine learning with web technologies, the Smart Lender project demonstrates how artificial intelligence can be applied to solve real-world problems in the banking and finance sector.

1.2 Purpose

The primary purpose of the Smart Lender – Loan Eligibility Prediction System is to automate the loan eligibility prediction process using machine learning. Instead of relying only on manual verification, the proposed system analyzes customer information and predicts the loan approval status quickly and accurately. This reduces processing time, minimizes human errors, and improves overall efficiency in the loan approval process.

Another important objective of this project is to demonstrate the practical implementation of machine learning in a real-world financial application. The system provides an easy-to-use interface where users can enter applicant details and instantly receive a prediction. It also helps financial institutions improve decision-making by providing data-driven recommendations based on historical loan records.

n addition, the project enhances learning in the fields of machine learning, web development, and data analysis by integrating various technologies into a single application. The Smart Lender system serves as a foundation for future enhancements, such as cloud deployment, integration with banking databases, improved prediction models, and secure user authentication, making it a scalable and practical solution for modern financial services.

Ideation Phase
Brainstorm & Idea Prioritization Template

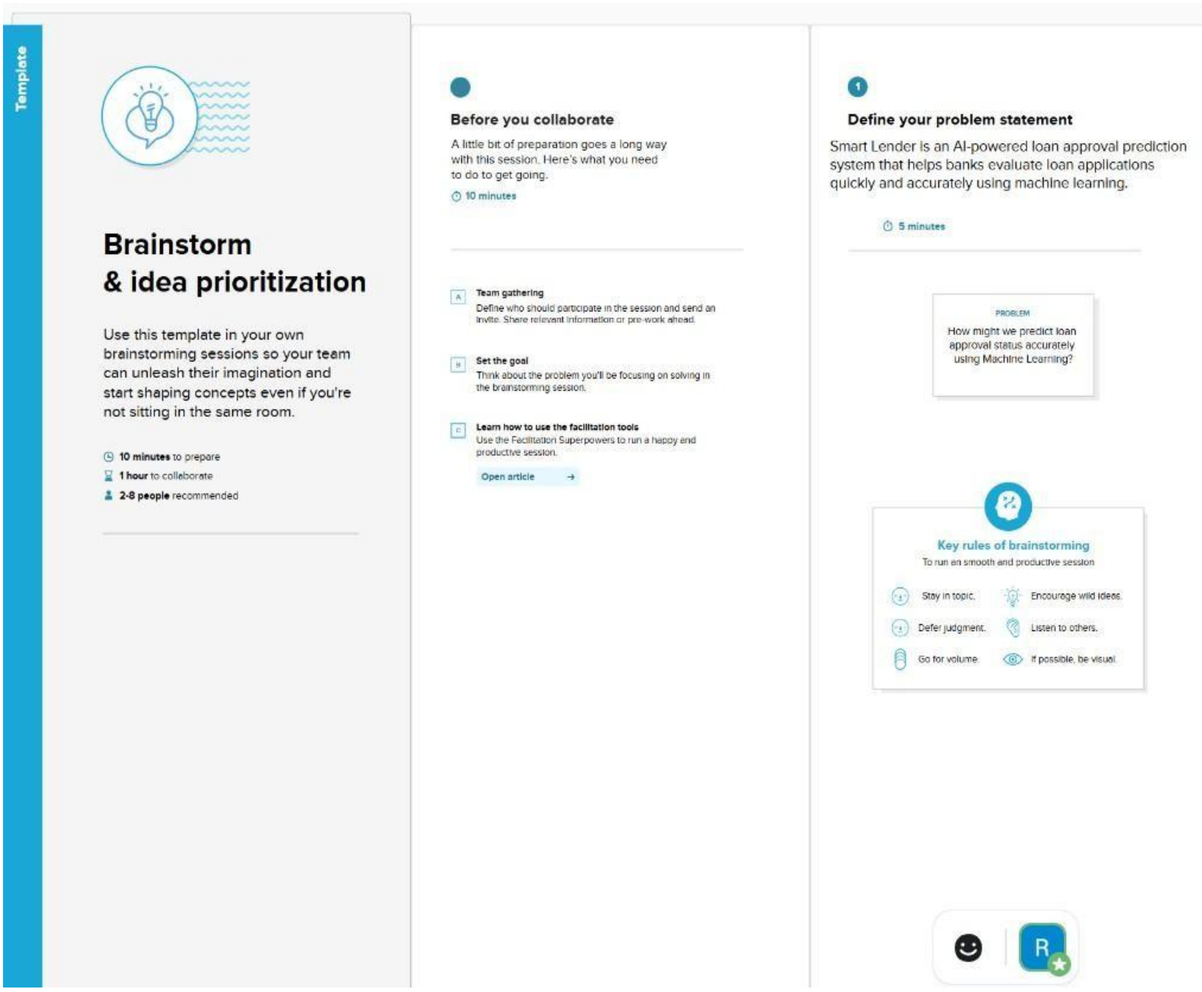
Date	30 June 2026
Team ID	SWUID2026211879 SWUID2026196379 SWUID2026211883 SWUID2026211881 SWUID2026226496
Project Name	**Smart Lender**
Maximum Marks	4 Marks

Brainstorm & Idea Prioritization Template:

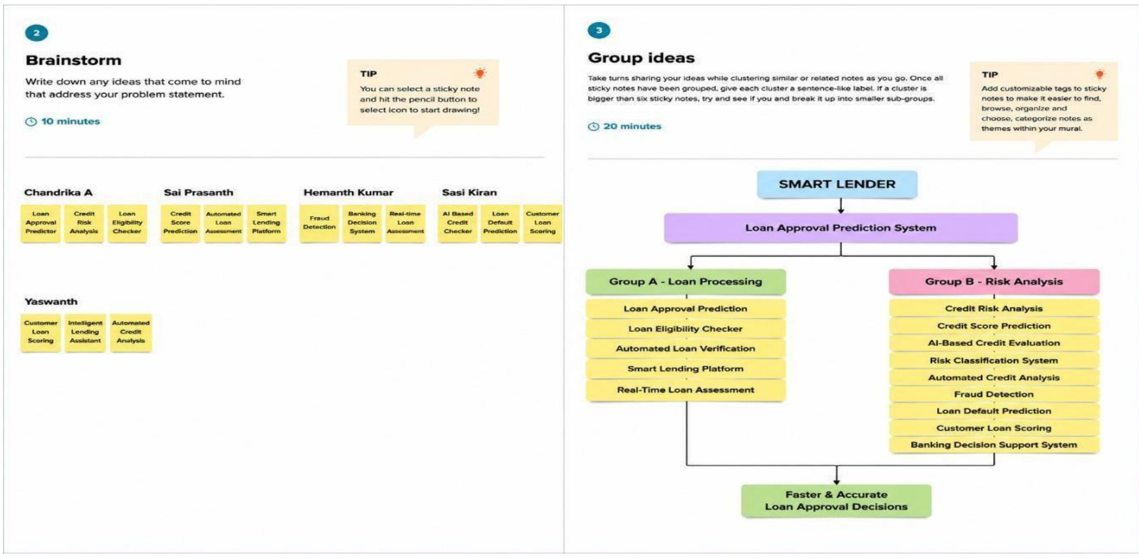
Smart Lender is a machine learning-powered loan approval prediction system designed to help banks and financial institutions make faster, more accurate, and data-driven lending decisions. The project encourages the use of advanced analytics and predictive modeling techniques to evaluate applicant creditworthiness based on factors such as income, credit history, employment status, education, and loan details. By leveraging machine learning algorithms and collaborative problem-solving approaches, the system aims to reduce loan default risks, improve approval efficiency, and support financial organizations in making reliable credit decisions.

Reference: https://www.mural.co/templates/brainstorm-and-idea-prioritization

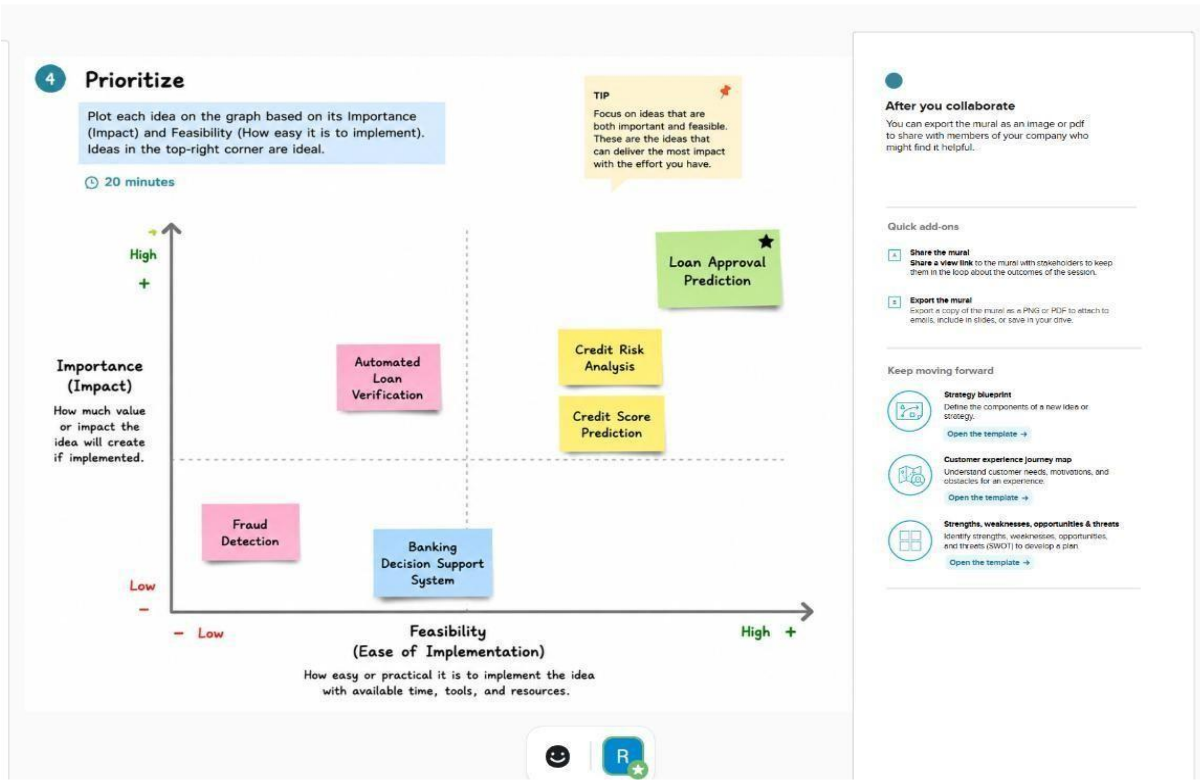
Step-1: Team Gathering, Collaboration and Select the Problem Statement



Step-2 Brainstrom ,Idea Listing and Grouping



Step-3: Idea Prioritization



Ideation Phase
Define the Problem Statements

Date	30 June 2026
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Project Name	Smart Lender
Maximum Marks	2 Marks

Customer Problem Statement Template:

Create a problem statement to understand your customer's point of view. The Customer Problem Statement template helps you focus on what matters to create experiences people will love.

A well-articulated customer problem statement allows you and your team to find the ideal solution for the challenges your customers face. Throughout the process, you'll also be able to empathize with your customers, which helps you better understand how they perceive your product or service.

I am	*Describe customer with 3-4 key characteristics - who are they?*	Loan applicants, salaried individuals, self-employed borrowers, small business owners who need financial support.
I'm trying to	*List their actions or goals - what are they trying to achieve?*	Get my loan approved quickly, easily and with the best possible terms.
but	*Describe what problems or barriers stand in the way - what stops them now?*	The loan process is complicated, requires lots of documents and takes a long time.
because	*Enter the "root cause" - why the problems or barriers exist - what needs to be solved?*	Manual verification, complex eligibility checks and lack of real-time data analysis cause delays and confusion.
which makes me feel	*Describe the emotions from the customer's point of view - how does it impact them?*	Frustrated, stressed and uncertain about loan approval status.

Reference: https://miro.com/templates/customer-problem-statement/

Example:

Problem Statement (PS)	I am (Customer)	I'm trying to	But	Because	Which makes me feel
PS-1	a loan applicant	get my loan approved quickly and easily	the approval process takes a long time	manual verification and document checking are time-consuming	frustrated and uncertain
PS-2	a bank loan approver	evaluate loan applications accurately and faster	there are too many applications and risk of human error	data is in different formats and verification is done manually	overwhelmed and pressured to make correct decisions

Ideation Phase
Empathize & Discover

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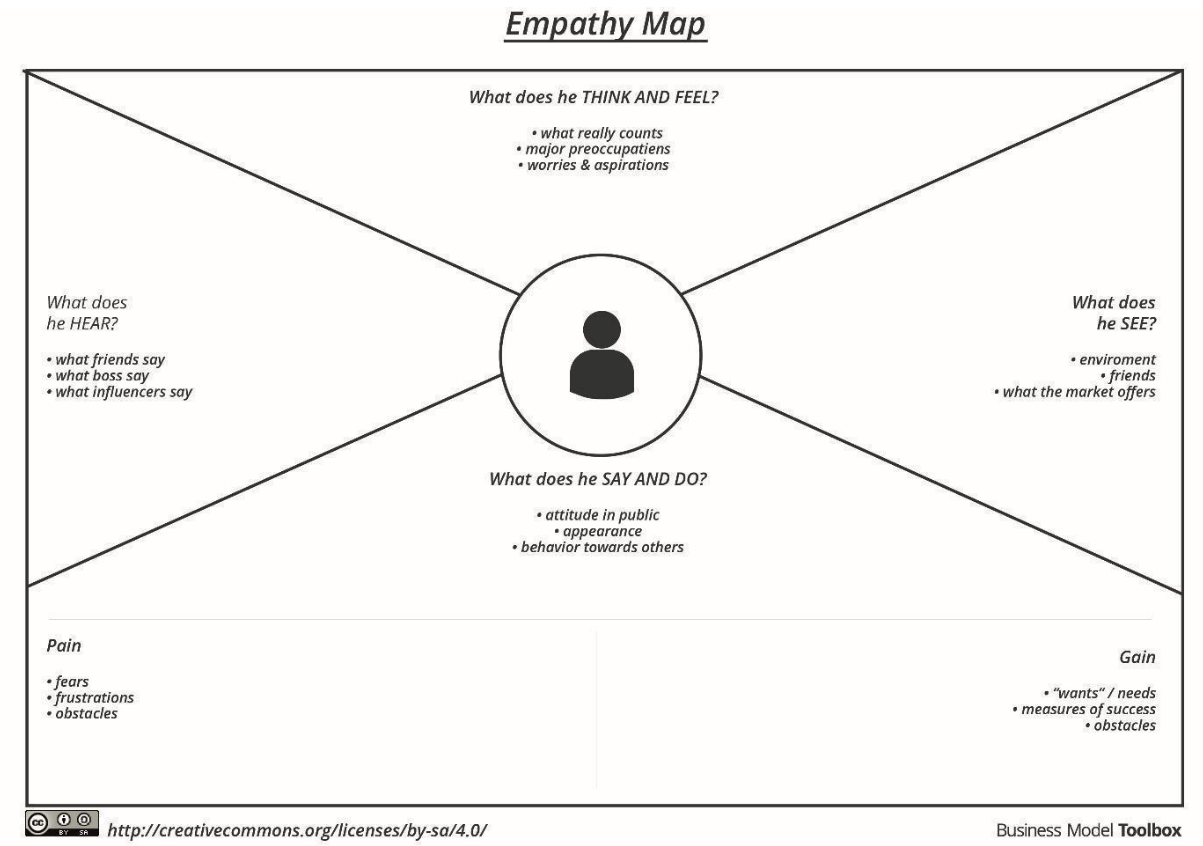
Empathy Map Canvas:

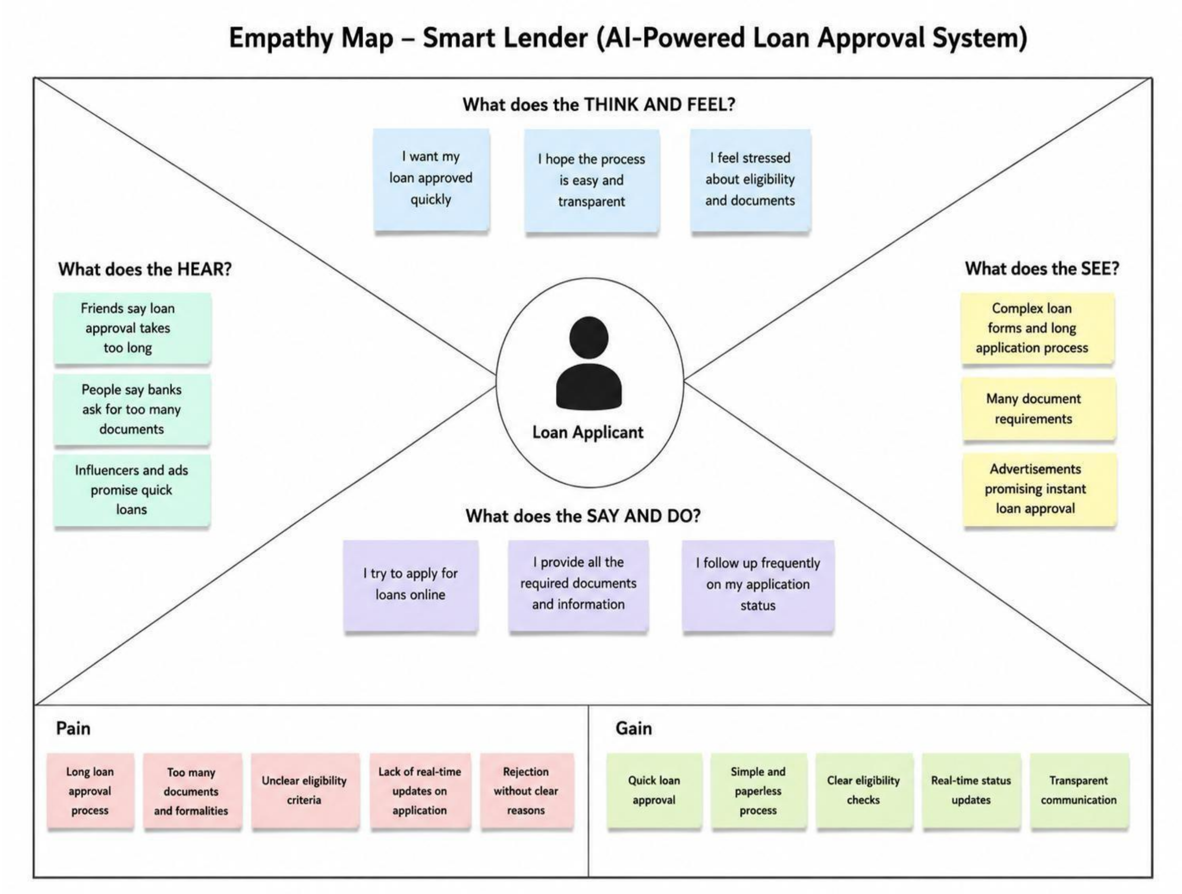
An empathy map is a simple, easy-to-digest visual that captures knowledge about a user's behaviours and attitudes.

It is a useful tool to helps teams better understand their users.
Creating an effective solution requires understanding the true problem and the person who is experiencing it. The exercise of creating the map helps participants consider things from the user's perspective along with his or her goals and challenges.

Example:



Reference: https://www.mural.co/templates/empathy-map-canvas



Empathy Map – Smart Lender (AI-Powered Loan Approval System)
What does the THINK AND FEEL?
I want my loan approved quickly
I hope the process is easy and transparent
I feel stressed about eligibility and documents
What does the HEAR?
Friends say loan approval takes too long
People say banks ask for too many documents
Influencers and ads promise quick loans
What does the SEE?
Complex loan forms and long application process
Many document requirements
Advertisements promising instant loan approval
Loan Applicant
What does the SAY AND DO?
I try to apply for loans online
I provide all the required documents and information
I follow up frequently on my application status
Pain
Long loan approval process
Too many documents and formalities
Unclear eligibility criteria
Lack of real-time updates on application
Rejection without clear reasons
Gain
Quick loan approval
Simple and paperless process
Clear eligibility checks
Real-time status updates
Transparent communication

Customer Journey Map

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Customer Journey Map

Map out the customer's experience stage-by-stage, capturing their actions, touchpoints, thoughts, and feelings, along with process ownership and improvement opportunities at each stage.

Phase of Journey	Stage 1 Awareness	Stage 2 Consideration	Stage 3 Decision
Actions What actions does the customer take?	• Learns about the loan through advertisement / social media • Visits website / app to know more	• Compares loan options and interest rates • Checks eligibility and required documents	• Fills the application form • Submits documents and applies
Touchpoint What are the customer's key touch points?	• Social Media Ads • Website • Friends / Referrals	• Website / App • Loan Calculator • Customer Support	• Application Form • Document Upload • Email / SMS Notifications
Customer Thoughts What is the customer thinking?	• Is this a reliable platform? • Will this loan be useful for me?	• Which loan is best for me? • What are the total charges?	• Is my application approved? • How quickly will I get the loan?
Customer Feeling How is the customer feeling?	• Curious • Cautious	• Interested • Need more clarity	• Hopeful • Anxious
Process Ownership Who is accountable at this stage?	• Marketing Team • Partnerships Team	• Product Team • Customer Support Team	• Operations Team • Credit Assessment Team
Opportunities Where can we improve the journey?	• Create more awareness campaigns • Improve website experience	• Simplify comparison tools • Provide clearer information	• Simplify application process • Reduce approval time

Project Design Phase-II
Data Flow Diagram & User Stories

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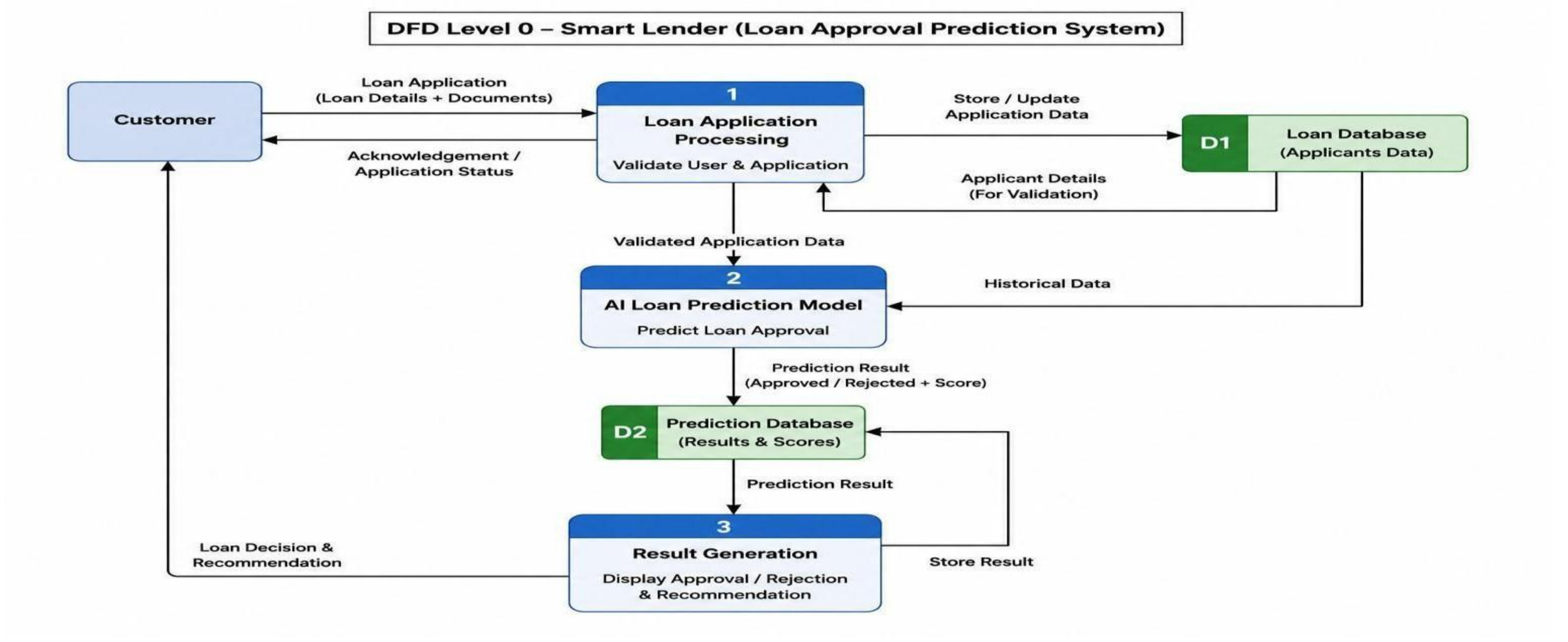
Data Flow Diagrams:

A Data Flow Diagram (DFD) is a traditional visual representation of the information flows within a system. A neat and clear DFD can depict the right amount of the system requirement graphically. It shows how data enters and leaves the system, what changes the information, and where data is stored.



User Stories

Use the below template to list all the user stories for the product.

User Type	Functional Requirement (Epic)	User Story Number	User Story / Task	Acceptance criteria	Priority	Release
Customer (Mobile User)	Registration	USN-1	As a user, I can register by entering my personal details and creating an account.	User account is created successfully.	High	Sprint-1
Customer (Mobile User)	Login	USN-2	As a user, I can log in using my email and password.	User can access the dashboard.	High	Sprint-1
Customer (Mobile User)	Loan Application	USN-3	As a user, I can fill the loan application form and upload the required documents.	Application is submitted successfully.	High	Sprint-1
Customer (Mobile User)	Loan Prediction	USN-4	As a user, I can check my loan approval prediction.	Prediction result is displayed.	High	Sprint-1
Customer (Mobile User)	Download Report	USN-5	As a user, I can download my prediction report.	Report is downloaded successfully.	Medium	Sprint-2
Customer (Web User)	Dashboard	USN-6	As a web user, I can view my application history and prediction results.	Dashboard displays complete records.	High	Sprint-2
Customer Care Executive	Application Support	USN-7	As a support executive, I can assist users with loan application issues.	Support requests are resolved.	Medium	Sprint-2
Administrator	User Management	USN-8	As an administrator, I can manage users and applications.	User and application records are updated.	High	Sprint-3
Administrator	AI Model Monitoring	USN-9	As an administrator, I can monitor AI prediction accuracy.	Performance reports are available.	Medium	Sprint-3
Administrator	Report Management	USN-10	As an administrator, I can generate system reports.	Reports are generated successfully.	Medium	Sprint-3

Project Design Phase-II
Solution Requirements (Functional & Non-functional)

Date	30 June 2026
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Maximum Marks	4 Marks

Functional Requirements:

Following are the functional requirements of the proposed solution.

FR No.	Functional Requirement (Epic)	Sub Requirement (Story / Sub-Task)
FR-1	User Registration	Registration through Form Registration through Gmail Registration through Linkedin
FR-2	User Confirmation	Confirmation via Email Confirmation via OTP
FR-3	Loan Application	Fill Loan Application Form, Upload Required Documents, Submit Application
FR-4	Loan Prediction	Validate Applicant Data, Predict Loan Approval using AI/ML, Display Prediction Result
FR-5	Dashboard	View Loan Status, View Application History, Download Prediction Report
FR-6	Profile Management	Update Personal Details, Change Password, Manage Account Settings

Non-functional Requirements:

Following are the non-functional requirements of the proposed solution.

FR No.	Non-Functional Requirement	Description
NFR-1	**Usability**	The application should provide a simple, user-friendly interface for all users.
NFR-2	**Security**	User data and loan information should be securely stored and encrypted.
NFR-3	**Reliability**	The system should provide accurate and consistent loan prediction results.

NFR-4	**Performance**	Loan prediction results should be generated within a few seconds.
NFR-5	**Availability**	The application should be available 24×7 with minimal downtime.
NFR-6	**Scalability**	The system should support multiple users and increasing application requests efficiently.

Project Design Phase-II
Technology Stack (Architecture & Stack)

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Maximum Marks	4 Marks

Technical Architecture:

The Deliverable shall include the architectural diagram as below and the information as per the table1 & table 2

Example: Order processing during pandemics for offline mode

Reference: https://developer.ibm.com/patterns/ai-powered-backend-system-for-order-processing-during-pandemics/

References:

https://c4model.com/

https://developer.ibm.com/patterns/online-order-processing-system-during-pandemic/

https://www.ibm.com/cloud/architecture

https://aws.amazon.com/architecture

https://medium.com/the-internal-startup/how-to-draw-useful-technical-architecture-diagrams-2d20c9fda90d

Table-1 : Components & Technologies:

S.No	Component	Description	Technology
1.	User Interface	Web interface for loan application and prediction.	HTML, CSS, JavaScript, React.js
2.	Application Logic-1	User registration, login and authentication.	Python (Flask)
3.	Application Logic-2	Loan application processing and validation.	Python, Pandas
4.	Application Logic-3	AI-based loan approval prediction.	Scikit-learn, NumPy
5.	Database	Store user, application and prediction details.	MySQL
6.	Cloud Database	Cloud-based database service.	Firebase / MySQL Cloud
7.	File Storage	Store uploaded documents and reports.	Local File System / Firebase Storage
8.	External API-1	Email notification service.	SMTP / Email API
9.	External API-2	OTP verification service.	Twilio OTP API
10.	Machine Learning Model	Loan Approval Prediction Model.	Random Forest Classifier (Scikit-learn)
11.	Infrastructure (Server / Cloud)	Application deployment on server / cloud.	Local Server / Render / Railway / AWS

Table-2: Application Characteristics:

S.No	Characteristics	Description	Technology
1.	Open-Source Frameworks	List the open-source frameworks used.	React.js, Flask, Scikit-learn
2.	Security Implementations	List all the security / access controls implemented, use of firewalls etc.	SHA-256, HTTPS
3.	Scalable Architecture	Justify the scalability of architecture (3 – tier, Micro-services).	Three-Tier Architecture
4.	Availability	Justify the availability of application (e.g. use of load balancers, distributed servers etc.).	Cloud Hosting
5.	Performance	Design consideration for the performance of the application (number of requests per sec, use of Cache, use of CDN's) etc.	Pandas, NumPy, Optimized ML Model

Project Design Phase
Problem – Solution Fit

Date	30 June 2026
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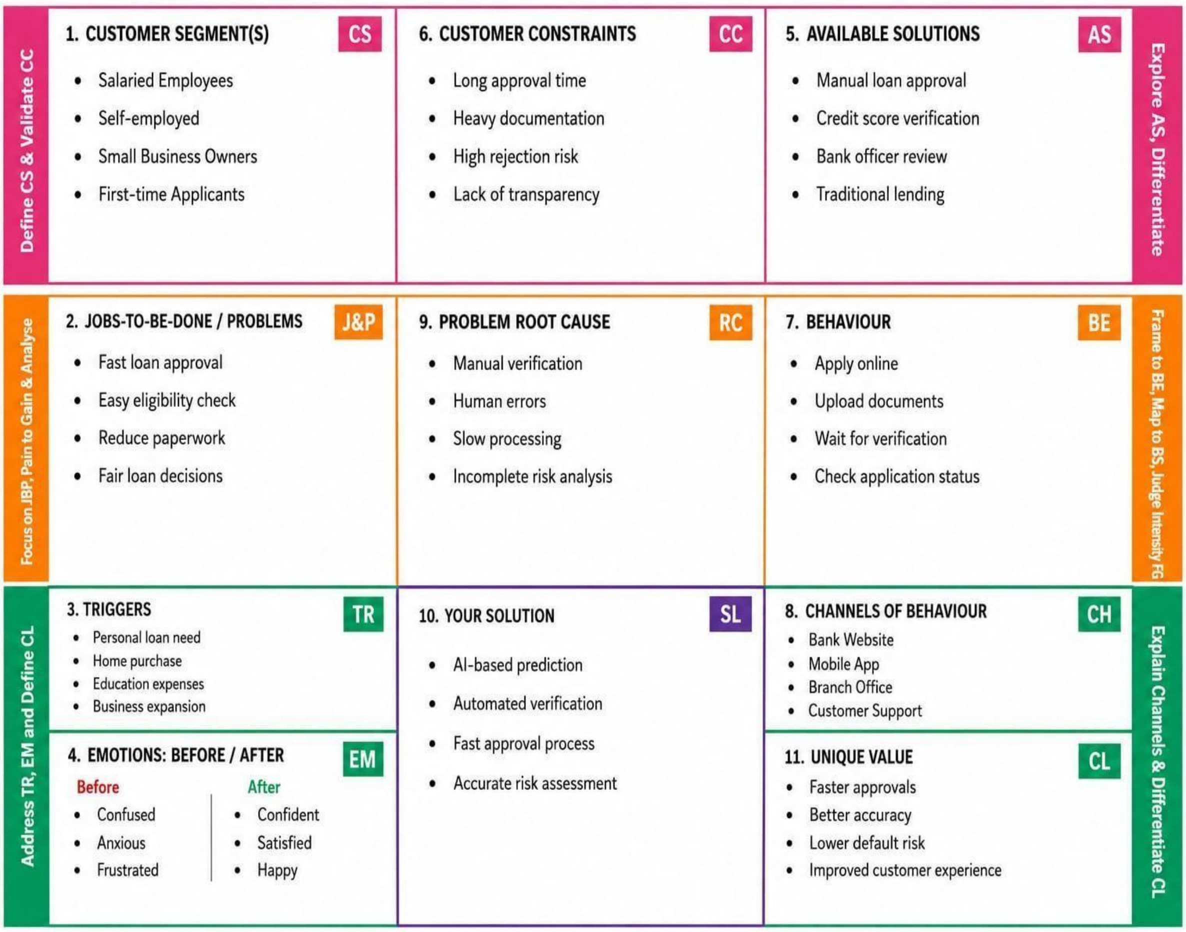
Project Name	Smart Lender
Maximum Marks	2 Marks

Problem – Solution Fit Template:

The Problem–Solution Fit means identifying the challenges faced by loan applicants and financial institutions, and providing a machine learning-based solution that accurately predicts loan approval. It helps banks and users make faster, consistent, and data-driven decisions while reducing manual effort and improving efficiency.

Purpose

- ❑ Predict loan approval quickly using a Machine Learning model..
- ❑ Reduce manual verification time and improve decision-making accuracy..
- ❑ Help banks process loan applications efficiently with reliable predictions..
- ❑ Improve customer experience by providing faster loan eligibility results..
- ❑ Understand applicant information effectively to support fair and informed loan approval decisions.



References:

1. https://www.ideahackers.network/problem-solution-fit-canvas/
2. https://medium.com/@epicantus/problem-solution-fit-canvas-aa3dd59cb4fe

Project Design Phase
Proposed Solution

Date	30 June 2026
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Project Name	Smart Lender
Maximum Marks	2 Marks

Proposed Solution:

Project team shall fill the following information in the proposed solution template.

S.No.	Parameter	Description
1.	Problem Statement (Problem to be solved)	Banks spend significant time reviewing loan applications manually, which can be slow and may lead to inconsistent decisions.
2.	Idea / Solution description	Develop a Machine Learning-based web application that predicts whether a loan application is likely to be approved based on applicant details.
3.	Novelty / Uniqueness	The system provides quick and accurate loan approval predictions through an easy-to-use web interface, reducing manual effort.
4.	Social Impact / Customer Satisfaction	It helps applicants receive faster responses and supports banks in making efficient, fair, and reliable loan decisions.
5.	Business Model (Revenue Model)	The solution can be offered to banks and financial institutions as a subscription-based or licensed software service.
6.	Scalability of the Solution	The application can be scaled to support more users, larger datasets, and additional financial services with minimal changes.

Project Design Phase
Solution Architecture

Date	30 June 2026
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Maximum Marks	4 Marks

Solution Architecture:

Solution architecture is the process of designing a Machine Learning-based web application that connects user inputs with a prediction model. It bridges the gap between loan application challenges and an intelligent software solution. Its goals are to:

- Develop a reliable Machine Learning solution for predicting loan approval.
- Describe the system architecture, workflow, and functionality to project stakeholders.
- Define the application features, development phases, and functional requirements.
- Provide a structured framework for developing, deploying, managing, and maintaining the loan prediction system.

Example - Solution Architecture Diagram:

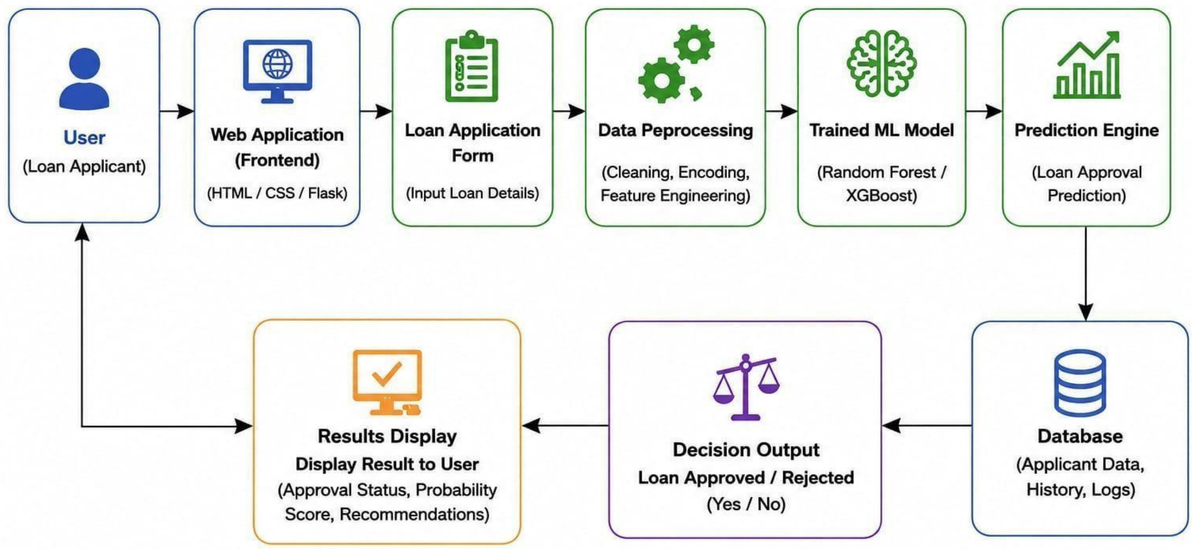

Figure 1: Architecture and data flow of the Smart Lender loan prediction system

Reference: https://aws.amazon.com/blogs/industries/voice-applications-in-clinical-research-powered-by-ai-on-aws-part-1-architecture-and-design-considerations/

Project Planning Phase

Project Planning Template (Product Backlog, Sprint Planning, Stories, Story points)

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Maximum Marks	5 Marks

Product Backlog, Sprint Schedule, and Estimation (4 Marks)

Use the below template to create product backlog and sprint schedule

Sprint	Functional Requirement (Epic)	User Story Number	User Story / Task	Story Points	Priority	Team Members
Sprint-1	Data Collection	USN-1	As a developer, I collect the loan dataset for model training.	2	High	Team
Sprint-1	Data Preprocessing	USN-2	As a developer, I clean and preprocess the dataset.	2	High	Team
Sprint-2	Model Training	USN-3	As a developer, I train the Machine Learning model for loan prediction.	3	High	Team
Sprint-2	Model Evaluation	USN-4	As a developer, I evaluate the model accuracy and performance.	2	Medium	Team
Sprint-3	Flask Integration	USN-5	As a developer, I integrate the trained model with the Flask application.	3	High	Team
Sprint-3	User Interface	USN-6	As a user, I can enter loan details through a simple web form.	2	High	Team
Sprint-4	Loan Prediction	USN-7	As a user, I can predict whether a loan is approved or rejected.	3	High	Team
Sprint-4	Testing & Deployment	USN-8	As a developer, I test and deploy the application successfully.	3	Medium	Team
	Dashboard					

Velocity:
Imagine we have a 10-day sprint duration, and the velocity of the team is 20 (points per sprint). Let's calculate the team's average velocity (AV) per iteration unit (story points per day)

$$AV = \frac{sprint\ duration}{velocity} = \frac{20}{10} = 2$$

Project Tracker, Velocity & Burndown Chart: (4 Marks)

Sprint	Total Story Points	Duration	Sprint Start Date	Sprint End Date (Planned)	Story Points Completed (as on Planned End Date)	Sprint Release Date (Actual)
Sprint-1	20	3 Days	24 Jun 2026	26 Jun 2026	20	26 Jun 2026
Sprint-2	20	2 Days	27 Jun 2026	28 Jun 2026	20	28 Jun 2026
Sprint-3	20	2 Days	29 Jun 2026	30 Jun 2026	20	30 Jun 2026
Sprint-4	20	3 Days	01 Jul 2026	03 Jul 2026	20	03 Jul 2026

Burndown Chart:

A burn down chart is a graphical representation of work left to do versus time. It is often used in agile software development methodologies such as Scrum. However, burn down charts can be applied to any project containing measurable progress over time.

https://www.visual-paradigm.com/scrum/scrum-burndown-chart/

https://www.atlassian.com/agile/tutorials/burndown-charts

Reference:

https://www.atlassian.com/agile/project-management

https://www.atlassian.com/agile/tutorials/how-to-do-scrum-with-jira-software

https://www.atlassian.com/agile/tutorials/epics

https://www.atlassian.com/agile/tutorials/sprints

https://www.atlassian.com/agile/project-management/estimation

https://www.atlassian.com/agile/tutorials/burndown-charts

Acceptance Testing
UAT Execution & Report Submission

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Maximum Marks	4 Marks

Purpose of Document

The purpose of this document is to present the testing process, test results, and defect analysis of the **Smart Lender – Applicant Credibility Prediction for Loan Approval** project before the final release. It summarizes the functional and performance testing performed on the application to ensure that the machine learning model and web application work correctly, reliably, and meet the expected requirements.

1. DEFECT ANALYSIS

This report shows the number of resolved or closed bugs at each severity level, and how they were resolved.

Resolution	Severity 1	Severity 2	Severity 3	Severity 4	Subtotal
By Design	10	4	2	3	20
Duplicate	1	0	3	0	4
External	2	3	0	1	6
Fixed	11	2	4	20	37
Not Reproduced	0	0	1	0	1
Skipped	0	0	1	1	2
Won't Fix	0	5	2	1	8
Totals	**24**	**14**	**13**	**26**	**77**

2. TEST CASE ANALYSIS

This report shows the number of test cases that have passed, failed, and untested.

Section	Total Cases	Not Tested	Fail	Pass
Print Engine	7	0	0	7
Client Application	51	0	0	51
Security	2	0	0	2
Outsource Shipping	3	0	0	3
Exception Reporting	9	0	0	9
Final Report Output	4	0	0	4
Version Control	2	0	0	2

Functional & Performance Testing

Model Performance Test

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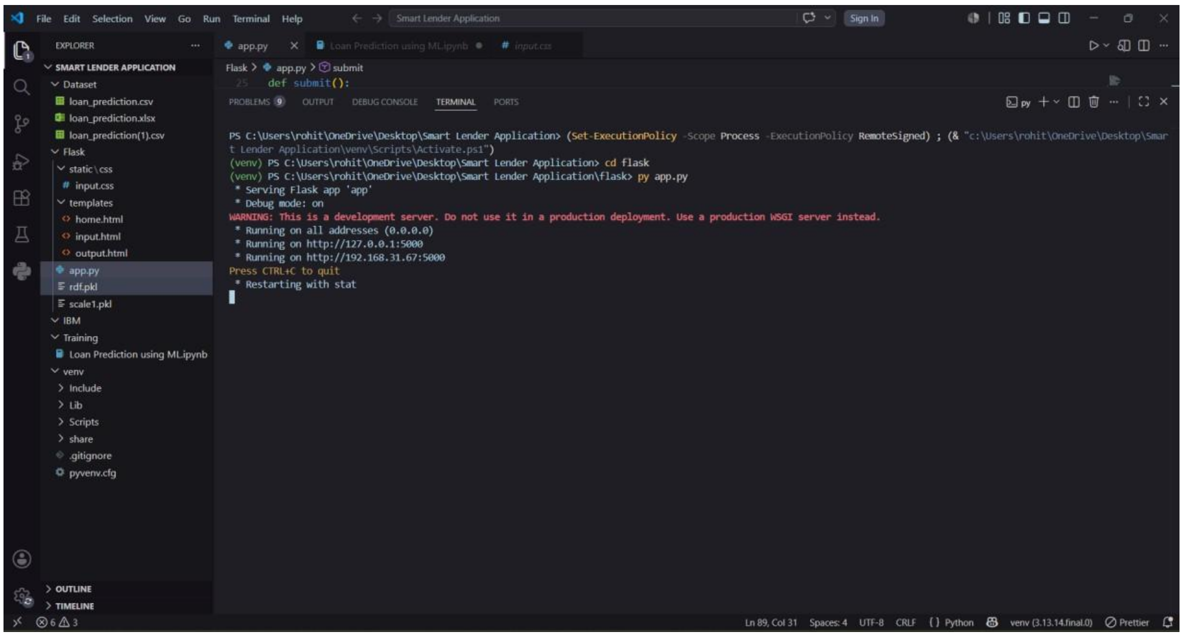
Maximum Marks	5

Test Scenarios & Results

Test Case ID	Scenario (What to test)	Test Steps (How to test)	Expected Result	Actual Result	Pass/Fail
FT-01	User Input Validation (e.g., required fields, invalid values)	Enter valid and invalid applicant details in input fields	Valid inputs are accepted, invalid inputs show an error message	Application accepted valid inputs and showed error for invalid inputs	**Pass**
FT-02	Loan Prediction (e.g., approve/reject prediction)	Enter complete applicant details and click "Predict" button	System predicts "Approved" or "Rejected" based on the input	Prediction displayed successfully as Approved or Rejected	**Pass**
FT-03	Model Integration Check (ML model with Flask application)	Submit applicant details to the application	Application sends input to the ML model and receives prediction	Model returned prediction successfully	**Pass**
FT-04	Application Navigation (e.g., Home, Predict pages)	Navigate between Home page and Prediction page	Pages load correctly without any errors	All pages loaded successfully	**Pass**
PT-01	Response Time Test	Submit applicant details and measure response time	Prediction should be displayed within 3 seconds	Prediction displayed within 3 seconds	**Pass**
PT-02	Multiple Prediction Test (API Speed Test)	Perform multiple predictions one after another	Application should respond correctly without slowing down	Multiple predictions completed successfully without slowdown	**Pass**
PT-03	Application Stability Test (e.g., long run, different inputs)	Run the application continuously and test with different inputs	Application should work without crashing or errors	Application remained stable throughout testing	**Pass**

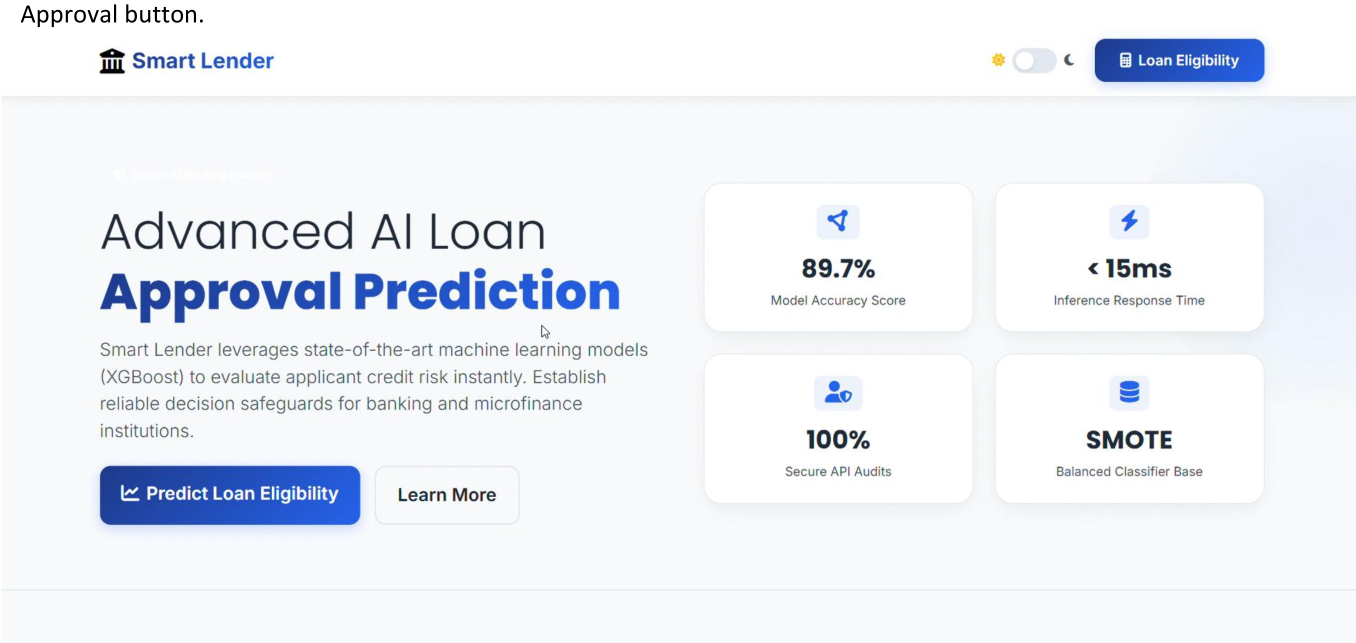
7.1 Output Screenshots

The Smart Lender – Loan Eligibility Prediction System was successfully developed and tested using a Flask-based web application. The application accepts user input such as applicant income, education, employment status, loan amount, credit history, and property area. Based on the entered details, the trained machine learning model predicts whether the loan application is likely to be approved or rejected.



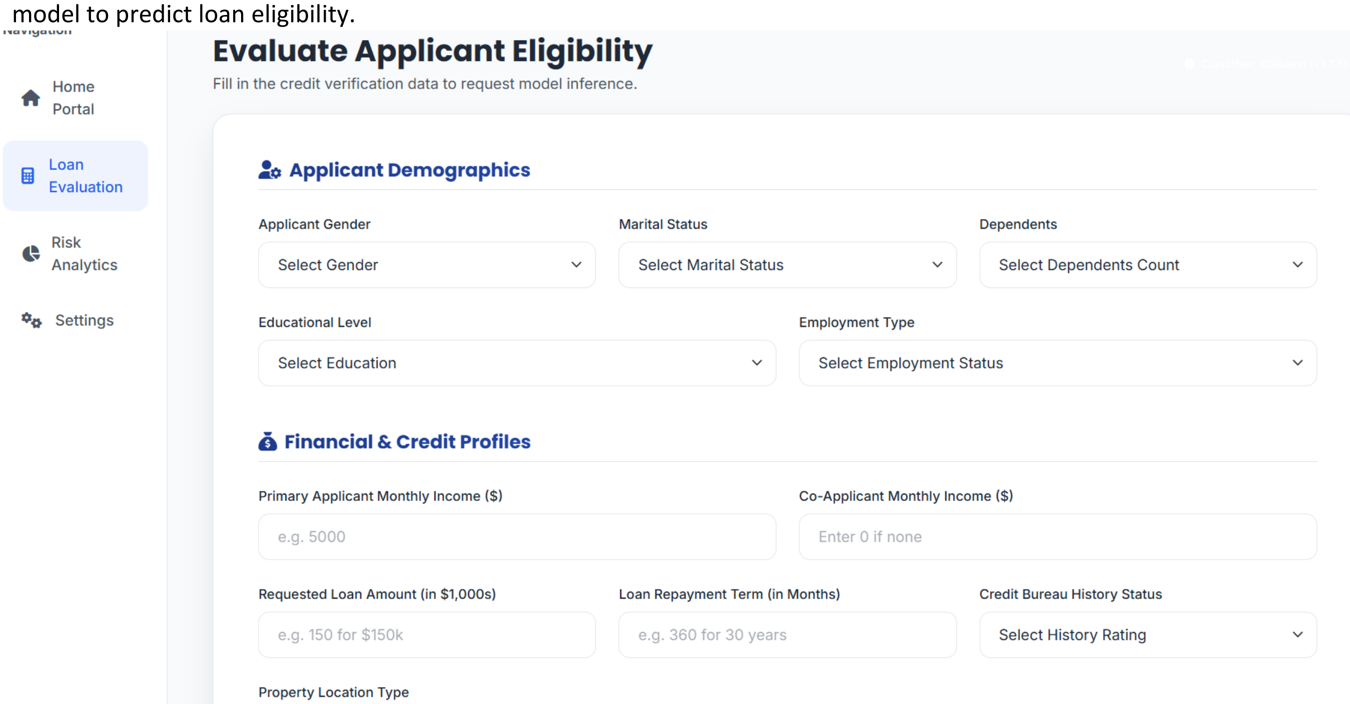
Home Page

The home page of the Smart Lender application provides an overview of the loan approval prediction system. It introduces the purpose of the application and allows users to navigate to the loan prediction form by clicking the Predict Loan Approval button.



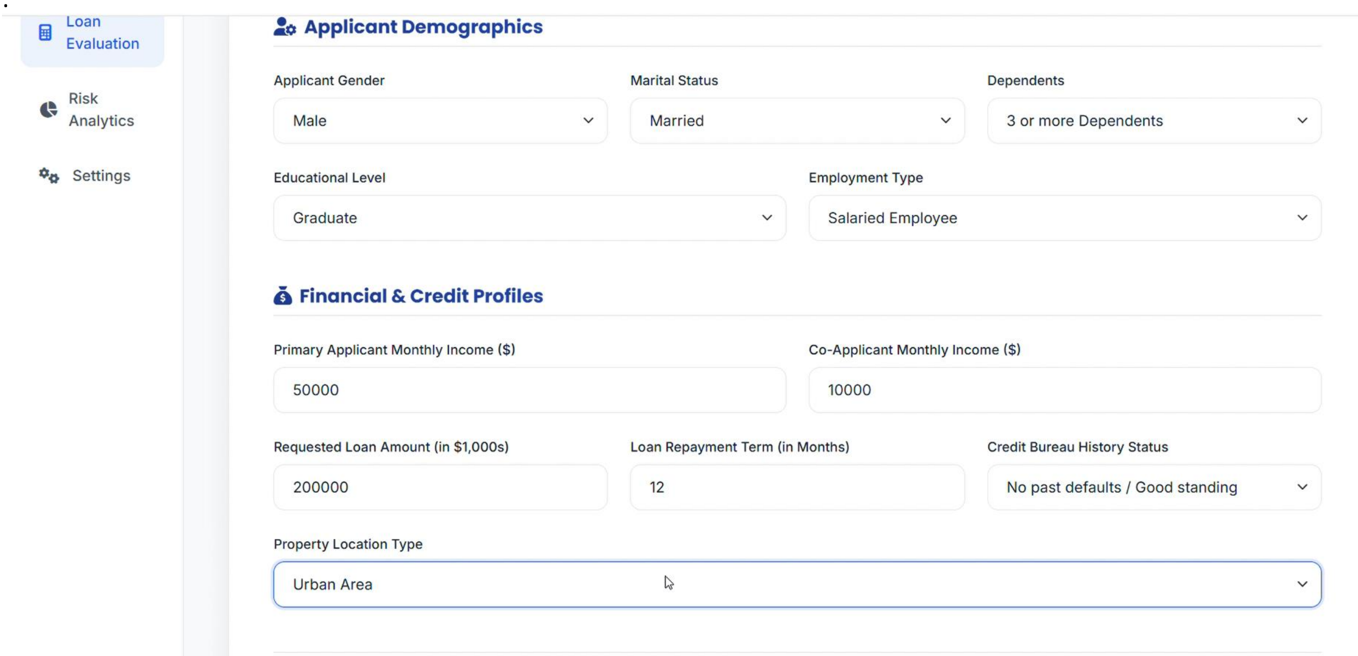
Loan Application Form

The loan application form allows users to enter applicant details such as gender, marital status, education, employment status, income, loan amount, loan term, credit history, and property area. These details are used by the machine learning model to predict loan eligibility.



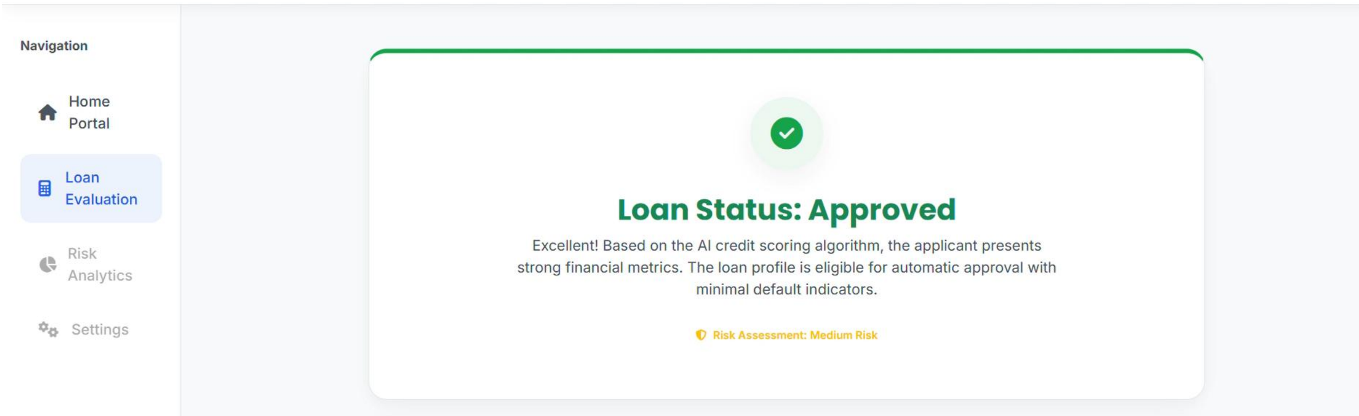
Loan Approval Prediction

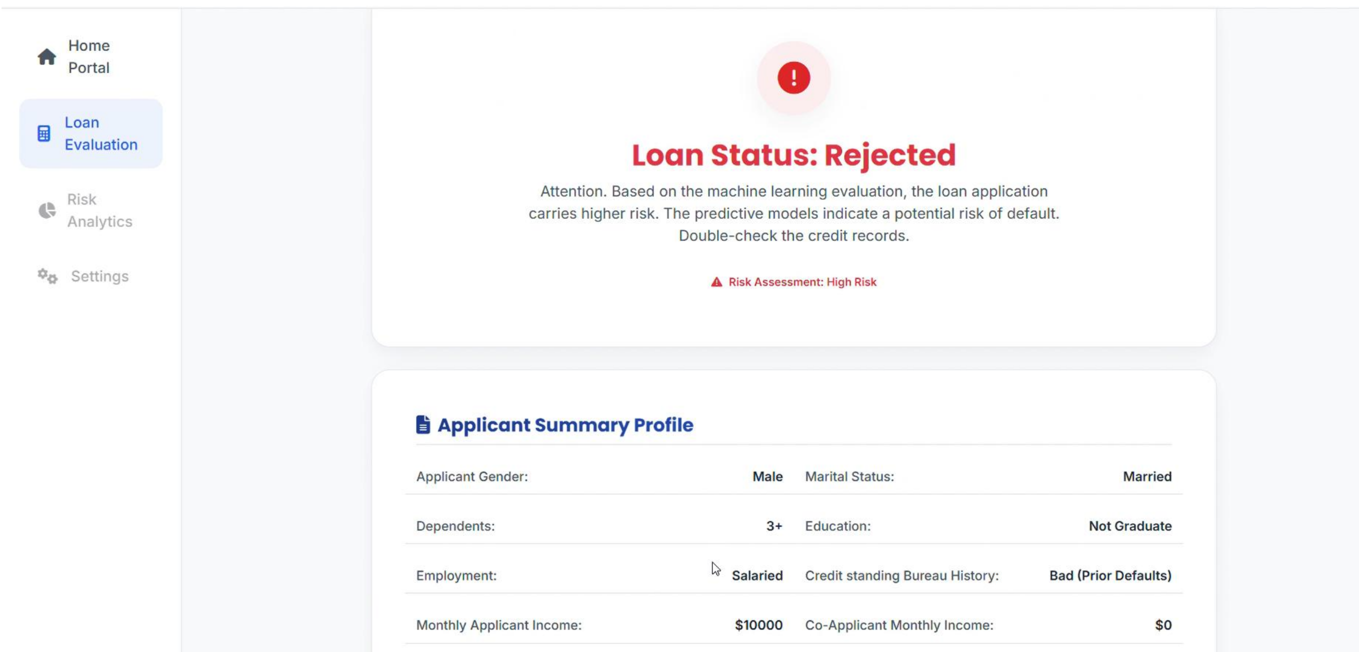
After submitting valid applicant details, the machine learning model analyzes the input and predicts the loan eligibility. In this case, the application successfully predicts that the loan is Approved and displays a confirmation message to the user
.



Prediction Result

The system processes the entered details and displays the loan approval prediction based on the trained machine learning model. This confirms that the application performs real-time loan eligibility prediction successfully





8. ADVANTAGES & DISADVANTAGES

Advantages

1. Provides fast and accurate loan eligibility prediction.
2. Reduces manual effort in the loan approval process.
3. Improves decision-making using machine learning.
4. Easy-to-use and user-friendly web interface.
5. Saves time for both customers and financial institutions.
6. Supports real-time loan prediction.

Disadvantages

- Prediction accuracy depends on the quality of the dataset.
- The model may require retraining when new data becomes available.
- Incorrect user input can affect prediction results.
- The system cannot completely replace final human verification in banks.

9. CONCLUSION

The Smart Lender – Loan Eligibility Prediction System was successfully developed and tested using machine learning and Flask. The application predicts loan eligibility based on applicant information and provides quick and reliable results. The project reduces manual effort, improves efficiency, and demonstrates the practical use of machine learning in the banking and finance sector.

10. FUTURE SCOPE

- Improve prediction accuracy using advanced machine learning models.
- Integrate the system with real-time banking databases.
- Add user authentication and role-based access.
- Deploy the application on cloud platforms.
- Develop a mobile application for easy access.
- Generate detailed loan recommendation reports.

11. APPENDIX

Source Code

The complete source code of the Smart Lender project is available in the GitHub repository.

GitHub Repository

https://github.com/NIKHITHARANGANA/Smart-Lender

Project Demo Link

https://drive.google.com/file/d/1tRcYK2Y5CWtLVFg

m2WXfuEOb9jPmndXq/view?usp=drive_link